

Genetic Analysis Report

Report ID:	SEQ-FE8A3E008914
Generated:	2025-12-04 17:32:28
Patient Name:	Genetic Risk Analysis Report - LOW RISK

Analysis Accuracy

93.7%

Patient Information

Analysis performed on file DNA-64233F993027. Processing completed using GENO AI v2.0 Elastic Net Logistic Regression model. Model trained on gene expression data with 94.2% cross-validation accuracy. Risk probability calculated: 26.0%

Variant Information

Gene Expression Analysis Complete • Analyzed Genes: 237 • Mean Expression Level: 33.49 • Expression Variance: 30991.88 • High Expression Markers: 55 • Risk Assessment: LOW RISK • Confidence: 26.0%

Analysis Results

{"report": {"sample_id": "SEQ-FE8A3E008914", "laboratory_user_id": "F55055", "generated_at": "2025-12-04T14:32:28.527419Z", "model_name": "geno_enet_pipeline.pkl", "model_version": "2.0", "total_genes_in_model": 237, "genes_analyzed": 237, "risk_level": "HIGH", "risk_score_percent": 99.43, "risk_score_raw": 0.9942849182079289, "risk_threshold": 75.0, "summary_title": "Genetic Risk Analysis Report - HIGH RISK", "summary_text": "The genetic analysis indicates a HIGH addiction risk profile based on the evaluated gene-expression markers. The model analyzed 237 genetic markers and calculated a risk probability of 99.43%. This score exceeds the 75% threshold, indicating elevated risk.", "top_genes": [{"gene": "ENSG00000140264.21", "expression": 2034.0}, {"gene": "ENSG00000159788.20", "expression": 1499.0}, {"gene": "ENSG00000125356.7", "expression": 599.0}, {"gene": "ENSG00000099795.7", "expression": 517.0}, {"gene": "ENSG00000100216.6", "expression": 376.0}, {"gene": "ENSG00000164604.14", "expression": 317.0}, {"gene": "ENSG0000015475.19", "expression": 282.0}, {"gene": "ENSG00000110092.4", "expression": 259.0}, {"gene": "ENSG00000150779.12", "expression": 241.0}, {"gene": "ENSG00000239382.11", "expression": 201.0}], "bottom_genes": [{"gene": "ENSG00000280027.1", "expression": 0.0}, {"gene": "ENSG00000283451.1", "expression": 0.0}, {"gene": "ENSG00000131007.9", "expression": 0.0}, {"gene": "ENSG00000255233.1", "expression": 0.0}, {"gene": "ENSG00000253809.1", "expression": 0.0}], "gene_stats": {"mean_expression": 33.4852, "median_expression": 0.0, "std_expression": 176.4177, "min_expression": 0.0, "max_expression": 2034.0, "genes_above_mean": 25,

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"validation": "Cross-validation"}}
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This report was generated automatically from genetic data analysis. For questions or concerns, please contact your healthcare provider.